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BATCH 45

LAB ASSESSMENT 4.3

Task 1: Zero-Shot Prompting – Leap Year Check

Scenario

Zero-shot prompting involves giving instructions without providing examples.

Task Description

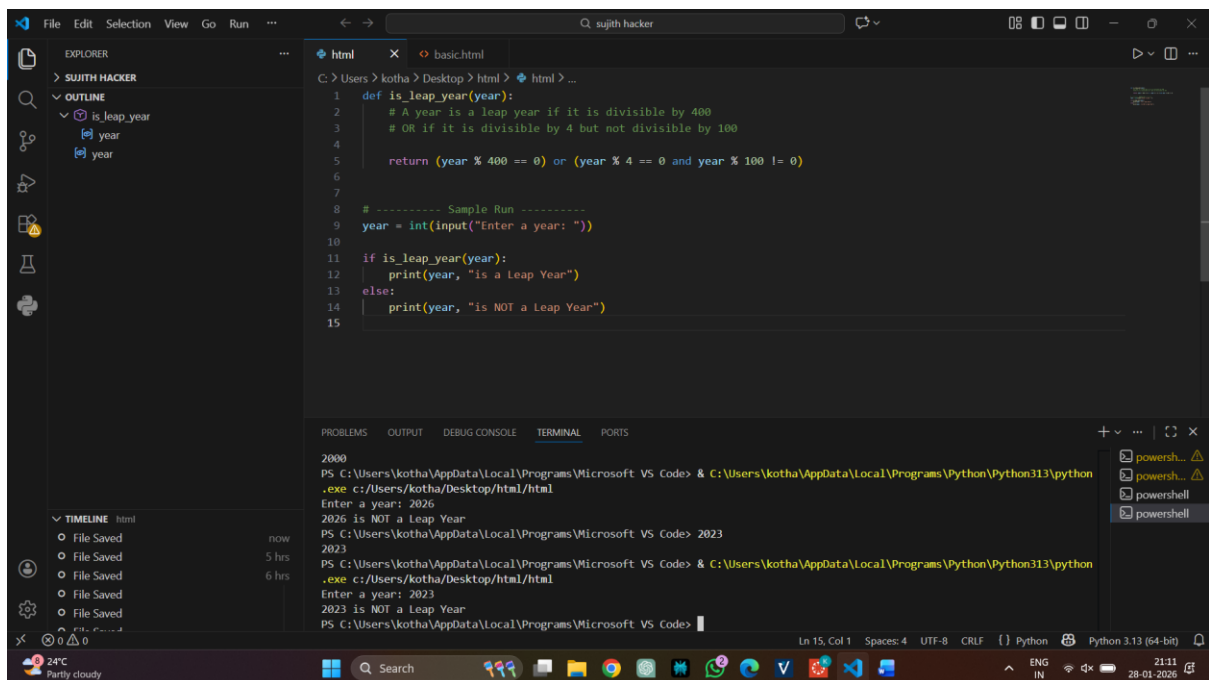
Use zero-shot prompting to instruct an AI tool to generate a Python function that:

- Accepts a year as input
- Checks whether the given year is a leap year
- Returns an appropriate result

Note: No input-output examples should be provided in the prompt.

Expected Output

- AI-generated leap year checking function
- Correct logical conditions
- Sample input and output
- Screenshot of AI-generated response (if required)



```
def is_leap_year(year):
    # A year is a leap year if it is divisible by 400
    # OR if it is divisible by 4 but not divisible by 100
    return (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0)

# ----- Sample Run -----
year = int(input("Enter a year: "))

if is_leap_year(year):
    print(year, "is a Leap Year")
else:
    print(year, "is NOT a Leap Year")
```

2000
PS C:\Users\kotha\AppData\Local\Programs\Microsoft VS Code> & C:\Users\kotha\AppData\Local\Programs\Python\Python313\python.exe c:/Users/kotha/Desktop/html/html
Enter a year: 2026
2026 is NOT a Leap Year
PS C:\Users\kotha\AppData\Local\Programs\Microsoft VS Code> 2023
2023
PS C:\Users\kotha\AppData\Local\Programs\Microsoft VS Code> & C:\Users\kotha\AppData\Local\Programs\Python\Python313\python.exe c:/Users/kotha/Desktop/html/html
Enter a year: 2023
2023 is NOT a Leap Year
PS C:\Users\kotha\AppData\Local\Programs\Microsoft VS Code>

Task 2: One-Shot Prompting – Centimeters to Inches Conversion

Scenario

One-shot prompting guides AI using a single example.

Task Description

Use one-shot prompting by providing one input-output example to generate a Python function that:

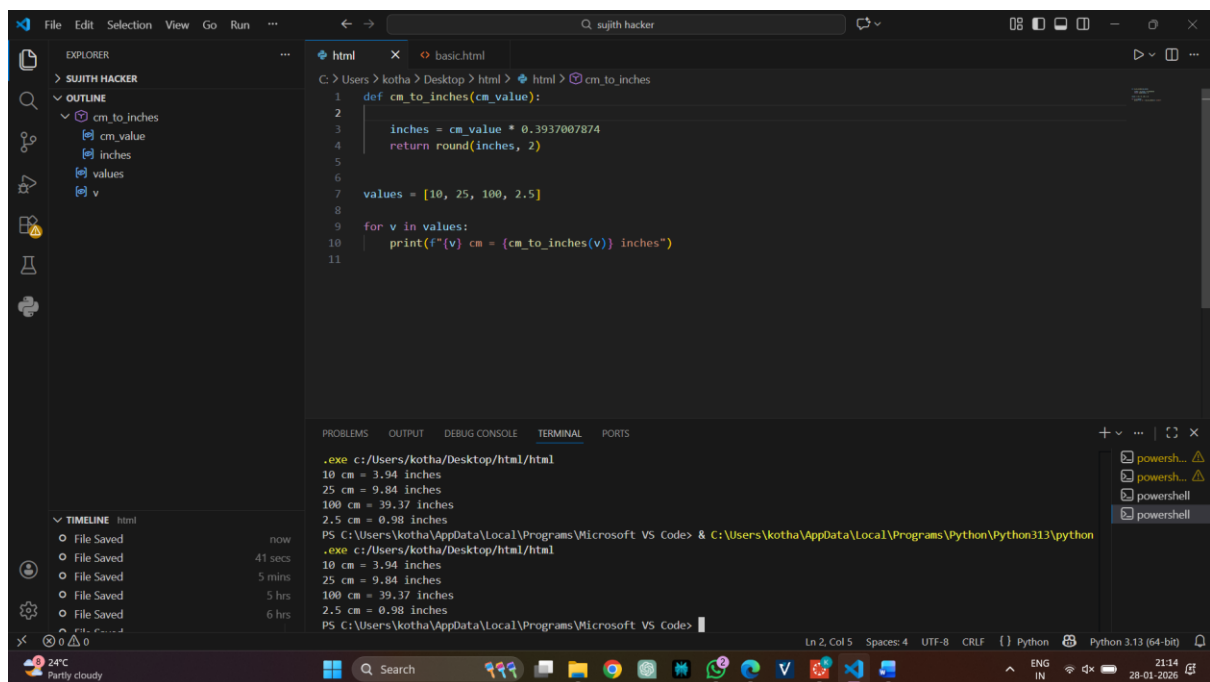
- Converts centimeters to inches
- Uses the correct mathematical formula

Example provided in prompt:

Input: 10 cm → Output: 3.94 inches

Expected Output

- Python function with correct conversion logic
- Accurate calculation
- Sample test cases and outputs



```
def cm_to_inches(cm_value):
    inches = cm_value * 0.3937007874
    return round(inches, 2)

values = [10, 25, 100, 2.5]
for v in values:
    print(f"{v} cm = {cm_to_inches(v)} inches")
```

```
.exe c:/Users/kotha/Desktop/html/html
10 cm = 3.94 inches
25 cm = 9.84 inches
100 cm = 39.37 inches
2.5 cm = 0.98 inches
PS C:/Users/kotha/AppData/Local/Programs/Microsoft VS Code> & C:/Users/kotha/AppData/Local/Programs/Python/Python313/python
.exe c:/Users/kotha/Desktop/html/html
10 cm = 3.94 inches
25 cm = 9.84 inches
100 cm = 39.37 inches
2.5 cm = 0.98 inches
PS C:/Users/kotha/AppData/Local/Programs/Microsoft VS Code>
```

Task 3: Few-Shot Prompting – Name Formatting

Scenario

Few-shot prompting improves accuracy by providing multiple examples.

Task Description

Use few-shot prompting with 2–3 examples to generate a Python function that:

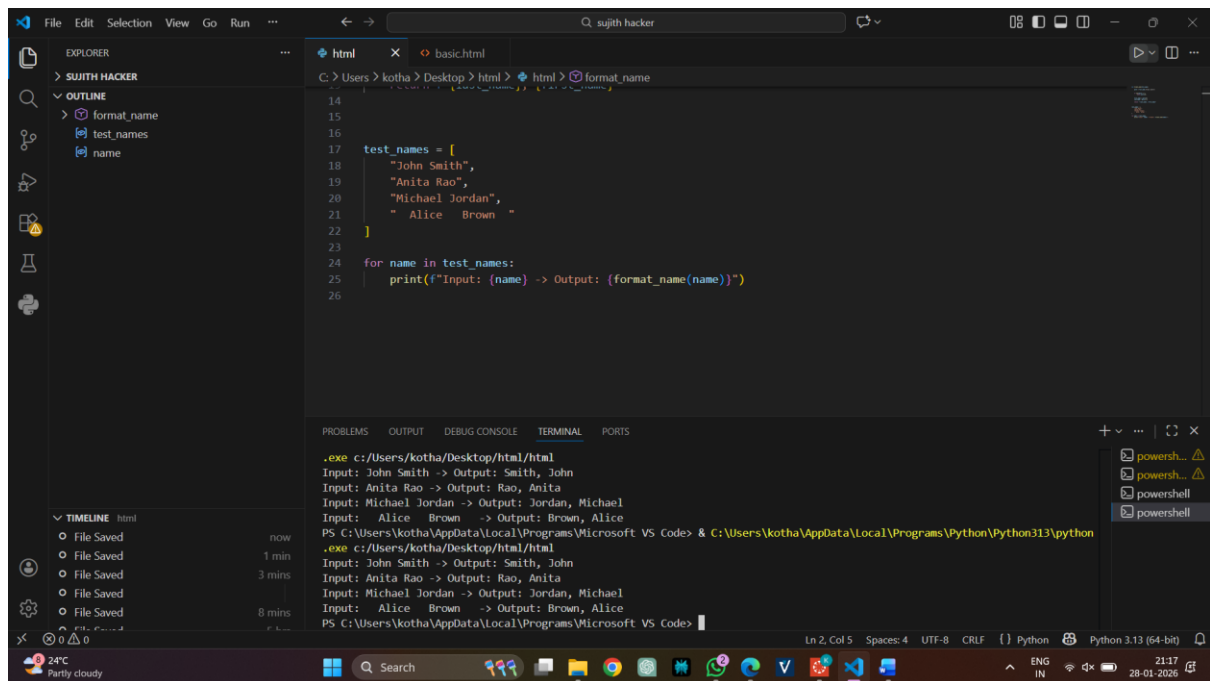
- Accepts a full name as input
- Formats it as “Last, First”

Example formats:

- "John Smith" → "Smith, John"
- "Anita Rao" → "Rao, Anita"

Expected Output

- Well-structured Python function
- Output strictly following example patterns
- Correct handling of names
- Sample inputs and outputs



The screenshot shows a Visual Studio Code editor window with a Python script in a file named `format_name.py`. The script defines a function `format_name` that takes a name as input and returns the name in the format 'Last, First'. It then uses a list `test_names` containing 'John Smith', 'Anita Rao', 'Michael Jordan', and 'Alice Brown' to test the function. The terminal output shows the results of running the script, displaying the input and output for each name in the list.

```
14
15
16
17 test_names = [
18     "John Smith",
19     "Anita Rao",
20     "Michael Jordan",
21     "Alice Brown"
22 ]
23
24 for name in test_names:
25     print(f"Input: {name} -> Output: {format_name(name)}")
26
```

Terminal Output:

```
.exe c:/Users/kotha/Desktop/html/html
Input: John Smith -> Output: Smith, John
Input: Anita Rao -> Output: Rao, Anita
Input: Michael Jordan -> Output: Jordan, Michael
Input: Alice Brown -> Output: Brown, Alice
PS C:\Users\kotha\AppData\Local\Programs\Microsoft VS Code> & C:\Users\kotha\AppData\Local\Programs\Python\Python313\python
.exe c:/Users/kotha/Desktop/html/html
Input: John Smith -> Output: Smith, John
Input: Anita Rao -> Output: Rao, Anita
Input: Michael Jordan -> Output: Jordan, Michael
Input: Alice Brown -> Output: Brown, Alice
PS C:\Users\kotha\AppData\Local\Programs\Microsoft VS Code>
```

Task 4: Comparative Analysis – Zero-Shot vs Few-Shot

Scenario

Different prompt strategies may produce different code quality.

Task Description

- Use zero-shot prompting to generate a function that counts vowels in a string
- Use few-shot prompting for the same problem
- Compare both outputs based on:
 - o Accuracy
 - o Readability
 - o Logical clarity

Expected Output

-
- The screenshot shows a Windows 10 desktop with a dark theme. The primary application is Visual Studio Code, which is open to a file named `basic.html`. The editor displays a Python script with the following code:
- ```

1 def count_vowels_zero(text):
2 vowels = "aeiouAEIOU"
3 total = 0
4
5 for ch in text:
6 if ch in vowels:
7 total += 1
8
9 return total
10

```
- The left sidebar of VS Code shows the Explorer view with a file named `count_vowels_zero` under an `OUTLINE` section. Below the Explorer is a Timeline view showing a list of file operations (File Saved) with timestamps (now, 2 mins, 3 mins, 5 mins).
- At the bottom of the VS Code window is a panel containing the TERMINAL view. The terminal shows the command prompt running the Python script, with the output being `5`. The terminal text is as follows:
- ```

PS C:\Users\kotha\AppData\Local\Programs\Microsoft VS Code> C:\Users\kotha\AppData\Local\Programs\Python\Python313\python
.exe c:/Users/kotha/Desktop/html/html
PS C:\Users\kotha\AppData\Local\Programs\Microsoft VS Code> C:\Users\kotha\AppData\Local\Programs\Python\Python313\python
.exe c:/Users/kotha/Desktop/html/html
PS C:\Users\kotha\AppData\Local\Programs\Microsoft VS Code> C:\Users\kotha\AppData\Local\Programs\Python\Python313\python
.exe c:/Users/kotha/Desktop/html/html
PS C:\Users\kotha\AppData\Local\Programs\Microsoft VS Code> C:\Users\kotha\AppData\Local\Programs\Python\Python313\python
.exe c:/Users/kotha/Desktop/html/html
PS C:\Users\kotha\AppData\Local\Programs\Microsoft VS Code> C:\Users\kotha\AppData\Local\Programs\Python\Python313\python
.exe c:/Users/kotha/Desktop/html/html
PS C:\Users\kotha\AppData\Local\Programs\Microsoft VS Code> 5

```
- The Windows taskbar at the bottom shows the system clock as 21:20 on 28-01-2026, along with various system icons and the Start button.

Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.

